# Smaller explicit Diophantine equations for ZFC consistency

## Encoding the monomials that occur, overlapping output positions, and polynomial approximations to square roots

AI-assisted mathematical research
Prepared for Vladimir Reshetnikov and the ProveIt project

23 October 2026

**Abstract**

We construct explicit integer polynomials whose unsolvability represents the syntactic consistency of ZFC, improving the current ProveIt constructions on both degree and arithmetic cost. The main nine-positive-variable equation has exact degree 15 534 796 and a division-free circuit with 655 live binary operations; its nine-nonnegative-variable version uses three more operations. A ten-unrestricted-integer equation has exact degree 39 914 856 and 651 operations. Lower-cost alternatives are supplied. The comparison is with the latest wave-6 constructions in repository commit `124c07c0b77a`, rather than a superseded baseline.

The first new observation is that a quadratic source need not distinguish all possible monomials: it must distinguish each monomial that occurs from all competing monomials. For the actual source this reduces the largest position from 799 to 179. A second finite certificate permits overlapping row blocks. A third improvement uses divisibility already present in the Jones arithmetic witnesses to construct a closer polynomial majorant for the radical in the final norm. All new arithmetic arguments are proved; the archive contains the literal circuits, complete fixed coefficient data, independent checks, and the inherited theorem dependencies. The resulting consistency equivalence is an ordinary mathematical proof formalizable in ZFC, using the same native compiler and machine calibration as the supplied repository. We do not claim a new end-to-end Lean/Rocq certificate, a proof of consistency, or a global minimum over Diophantine equations.

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# 1 The result, its domain, and the comparison

Write $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$. We use the repository's precise syntactic consistency sentence

$$\mathrm{Con}(\mathrm{ZFC}) \;\equiv\; \neg\exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot),$$

calibrated there by `BProv ZFCax_s [] Form.fBot`. The coefficient tuple is fixed independently of this sentence's truth.

**Theorem 1.1** (Explicit consistency equations)**.** *The finite data and polynomial formulas in this article define polynomials $F_9 \in \mathbb{Z}[x_1, \ldots, x_9]$ and $F_{10} \in \mathbb{Z}[z_1, \ldots, z_{10}]$ such that, using the inherited mathematical compiler and arithmetic theorems identified in Section 7, ZFC proves*

$$\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{9}\ F_9(\mathbf{x}) = 0 \;\longleftrightarrow\; \mathrm{Con}(\mathrm{ZFC}),$$
$$\neg\exists \mathbf{z} \in \mathbb{Z}^{10}\ F_{10}(\mathbf{z}) = 0 \;\longleftrightarrow\; \mathrm{Con}(\mathrm{ZFC}).$$

*The principal $F_9$ has exact total degree 15 534 796 and 655 binary operations. The principal $F_{10}$ has exact total degree 39 914 856 and 651 operations. The displayed registers in their definitions are abbreviations, not additional unknowns.*

| Construction | Domain | Exact degree | Operations |
|---|---|---|---|
| Package 45 (previous) | $\mathbb{Z}_{>0}^{9}$ | 129 875 980 | 788 |
| New source, uniform, inherited tail | $\mathbb{Z}_{>0}^{9}$ | 29 157 244 | 562 |
| New source, uniform, V8 | $\mathbb{Z}_{>0}^{9}$ | 24 399 300 | 568 |
| New source, uniform, principal tail | $\mathbb{Z}_{>0}^{9}$ | 20 754 916 | 599 |
| New source, overlap, V8 | $\mathbb{Z}_{>0}^{9}$ | 18 262 476 | 624 |
| **New source, overlap, principal** | $\mathbb{Z}_{>0}^{\mathbf{9}}$ | **15 534 796** | **655** |
| Package 40 (previous) | $\mathbb{Z}^{10}$ | 237 520 456 | 782 |
| New source, uniform | $\mathbb{Z}^{10}$ | 53 325 896 | 595 |
| **New source, overlap** | $\mathbb{Z}^{\mathbf{10}}$ | **39 914 856** | **651** |

Table 1: The new degree–operation tradeoffs. Every nine-variable natural-domain version has the same degree and three additional operations. The archive also includes the overlapping source with the inherited tail (618 operations, degree 21 823 612). Certificate-circuit operations are not charged to the primary equation.

Relative to the latest comparator, the principal nine-variable degree is smaller by a factor of approximately 8.3603, and its operation count falls by 133 (16.88%). The principal ten-integer degree is smaller by a factor of approximately 5.9507, and its operation count falls by 131 (16.75%).

The nine-variable result retains positivity. Its unrestricted-integer interpretation is not asserted. The supplied natural-variable version uses an explicit coordinate shift described below. For a Diophantine equation over all integers, $F_{10}$ is the corresponding primary deliverable. That branch retains Sun's ten-integer reduction [7]; the nine-variable branch uses the older Jones arithmetic bridge [6].

The preceding nine-positive-variable point in package 45 is degree 129 875 980 at 788 operations. The preceding ten-integer point in package 40 is degree 237 520 456 at 782 operations [2, 3]. We compare against those latest points. Earlier claims comparing only with exponentially larger wave-5 polynomials are not the baseline for this contribution.

There is no metric-independent "simplest" equation. The operation count charges each binary addition, subtraction, and multiplication, including multiplication by a fixed coefficient. Reuse of a computed

register is free; fixed powers are expanded into charged multiplication gates. The count does not measure bit complexity, expanded monomial count, or the cost of producing fixed coefficients. The repository's 67-operation equation with fifteen positive variables remains cheaper on that single measure. We have not found a sub-nine-natural-variable or sub-ten-integer construction.

## 2 The fixed quadratic parent and coefficient data

The common parent has 31 strictly positive coordinates and 22 quadratic equations. In order, the coordinates are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta.$$

Here these letters are local to the parent; later tail registers reuse some letters for different quantities. The four fixed compiler ports are $(\beta, K_0, M_C, M_F)$, with the high mask $M_F$ already shifted. Put

$$q = \beta J + 1, \qquad k = \eta + \zeta, \qquad H = 4a + 3, \qquad C = q - F - \alpha.$$

The equations, in their original order, are

| Row | Equation |
|---|---|
| 1 | $L + 1 = q^2$ |
| 2 | $N = q(L + 1)$ |
| 3 | $X = wq$ |
| 4 | $a - E = sN$ |
| 5 | $E = X(a - E)$ |
| 6 | $c - \eta = k(a - E)$ |
| 7 | $P - k = E(c - \eta)$ |
| 8 | $\tau^2 = P(P - k) + 1$ |
| 9 | $D = X + ac + \gamma H$ |
| 10 | $\Delta = a^2 + H$ |
| 11 | $C_2 = c^2$ |
| 12 | $D^2 = \Delta C_2 + 1$ |
| 13 | $k = jc - V + 1 + hE$ |
| 14 | $jc - V = (q - F)(N - q) - CL + (M_C + qM_F)J$ |
| 15 | $(K_0 + w)C = F + z\beta J$ |
| 16 | $F_2 = f^2$ |
| 17 | $V = of - c$ |
| 18 | $g = y^2 - V^2$ |
| 19 | $I = iC_2$ |
| 20 | $B_a = I^2$ |
| 21 | $B_a = \Delta(F_2 - 1)$ |
| 22 | $y^2 - B_a g = 1$ |

The exact orientation of each residual is fixed by `data/parent_system.json`. The displayed equalities alone are unchanged by reversing a residual; the finite coefficient file records the orientation used in the packing. For each parent coordinate substitute $x_i + 1$ and form the recorded left-minus-right residual. Denote these polynomials by $r_1, \ldots, r_{22}$. Every $r_j$ has degree at most two. The fifth residual has $r_5(\mathbf{0}) = 1$.

The imported parent theorem is

$$\exists \mathbf{x} \in \mathbb{N}^{31} \ \bigwedge_{j=1}^{22} r_j(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \exists p \ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot). \tag{1}$$

We use the complete package-31 tuple, whose radix is $B_0 = 2^{1064921}$. Its four port bit lengths are respectively

$$1064921, \quad 1070004, \quad 1064921, \quad 1064922.$$

Their compressed unsigned little-endian payloads are included, together with hashes and the inherited coefficient recipe. They total 533,099 uncompressed bytes. No component is taken from a different compiler tuple. The archive also materializes all 16 fixed leaves of the emitted graphs in `data/fixed/`, including signs, exact byte lengths, and SHA-256 hashes; these payloads contain 3,331,623 uncompressed bytes in total.

Every additional fixed leaf is either a literal small integer, the explicit integer $2^{2140030}$, or a recorded integer polynomial in these four ports. Thus the equation contains no unspecified recursively enumerable index, consistency-dependent coefficient, or search for a ZFC refutation in its definition. A gate file plus these finite leaves is a complete specification of one ordinary integer polynomial, even though fully expanding it would be an inefficient way to display it.

## 3 A smaller positional certificate

### 3.1 Only the present monomials need isolation

Adjoin $x_0 = 1$ and homogenize each monomial of degree at most two to degree two. There are $32 \cdot 33/2 = 528$ possible unordered pairs of indices. Only 71 of those pairs occur in the 22 residuals. We call these pairs *present*. Assign the following weights in the coordinate order $(0, J, F, \ldots, \Delta)$:

$$\begin{gathered}(0, 174, 166, 110, 137, 115, 65, 168, 126, 105, 49, 155, 54, 60, 15, 4,\\ 118, 146, 26, 44, 7, 17, 57, 38, 158, 177, 179, 116, 46, 94, 35, 96).\end{gathered} \tag{2}$$

Let $W = 179$. All nonpadding weights are distinct and positive. For every present pair $(a, b)$, its sum $w_a + w_b$ differs from the sum of every other pair among the 528. Pairs absent from every residual may collide. Indeed, these 528 pairs occupy just 266 different positions.

The certificate is a finite list and a universally quantified check over 528 pairs, not a claim of optimality for a combinatorial search. Both the construction verifier and a separately written sparse-polynomial checker expand the actual parent to identify the 71 present pairs and check this property. No heuristic search code is needed to validate it.

### 3.2 Extraction with possibly overlapping row blocks

Write $r_j = \sum_\alpha p_{j,\alpha} x^\alpha$, where $|\alpha| \le 2$, and set

$$s(\alpha) = \sum_i \alpha_i w_i, \qquad c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i! \, p_{j,\alpha}.$$

For a row permutation $\pi$, signs $\sigma_j \in \{1, -1\}$, and increasing integer targets $H_j \ge 2W$, define

$$\mathcal{D}(T) = \sum_{j=0}^{21} \sum_{\alpha \in \mathrm{supp}\, r_{\pi(j)}} \sigma_j c_{\pi(j),\alpha} T^{H_j - s(\alpha)}. \tag{3}$$

Every exponent here is nonnegative and fixed.

**Lemma 3.1** (Isolation only at the selected outputs)**.** *Suppose for every row $j$, every $\alpha \in \operatorname{supp} r_{\pi(j)}$, every degree-at-most-two monomial $\beta$, and every row $i$, the equality*

$$H_i - H_j = s(\beta) - s(\alpha)$$

*implies $i = j$ and $\beta = \alpha$. Then, for $\mathcal{C}(T) = (1 + \sum_i x_i T^{w_i})^2$,*

$$[T^{H_j}]\,\mathcal{C}(T)\mathcal{D}(T) = 2\sigma_j r_{\pi(j)}(\mathbf{x}).$$

*Moreover all monomial records in $\mathcal{D}$ have distinct exponents.*

*Proof.* A term from row $j$ and monomial $\alpha$ can contribute at output $H_i$ only under the displayed equality. The isolation hypothesis leaves exactly the matching monomial in the matching row. Its coefficient in $\mathcal{C}$ is $2!\,x^\alpha/((2-|\alpha|)!\prod_i \alpha_i!)$, cancelling the factorial multiplier in $c_{j,\alpha}$. This argument also treats constants and linear terms by the padding variable. If two records of $\mathcal{D}$ shared an exponent, multiplying either by its own present monomial from $\mathcal{C}$ would violate output isolation at that row. Hence the records have distinct exponents. □

The condition distinguishes a present monomial from *every* competing monomial. Distinguishing present monomials only from one another would not be sufficient: an absent monomial is still produced by $\mathcal{C}$ and could contaminate a selected coefficient.

### 3.3 Two explicit placements

The uniform placement uses row order $1, 2, 3, 4, 6, \ldots, 22, 5$, all signs positive, and

$$H_j = 358 + 359j \quad (0 \le j \le 21), \qquad H_{21} = 7897.$$

Distinct row targets are more than $2W$ apart, so the 71-pair certificate proves the lemma's hypothesis immediately. The highest coefficient of $\mathcal{D}$ is $2r_5(\mathbf{0}) = 2$.

The overlapping placement has targets

$$\begin{aligned}(&358, 608, 939, 1147, 1432, 1739, 1989, 2170, 2512, 2699, 2972,\\ &3305, 3492, 3683, 3914, 4108, 4399, 4697, 4926, 5224, 5469, 5775).\end{aligned} \tag{4}$$

Its original row order and signs are specified in `data/overlap_layout.json`; the finite table is reproduced in the appendix. The last row is the negative of original row 9 and has constant $+8$, so the highest coefficient of $\mathcal{D}$ is 16. Every successive gap is greater than $W$. Differences of nonadjacent targets therefore exceed $2W$, and only adjacent pairs require the finite cross-row check. That check verifies the full hypothesis of Lemma 3.1. No claim that all support intervals are disjoint is made or needed.

## 4 From isolated coefficients to the arithmetic source

We spell out the source because replacing positions inside a universal theorem with hard-coded exponents would be invalid. The new source is proved directly and then supplied to the numerical arithmetic bridge.

Let $H_* = \max_j H_j$, $K_* = H_* + 2W$, and let $M_*$ bound the absolute values of every weighted coefficient $c_{j,\alpha}$. The finite coefficient check gives $M_* < 2^{2134927}$. The chosen fixed integer $\Lambda = 2^{2140030}$ comfortably satisfies all coefficient bounds below. For the nine-variable construction use

$$b = \Lambda + \epsilon, \quad B = b^3, \quad \epsilon \geq 0;$$

for the ten-integer construction use $b = 1 + 3f$, $B = \Lambda b^2$. The latter bridge first recovers $b \geq 4$ before the digit argument is used.

Define the following polynomial registers, with $g$ the encoded witness:

$$\ell = \sum_{i=1}^{31} B^{w_i}, \qquad N_0 = B^{W+1}, \qquad N_1 = 4B^{K_*+1}, \quad N = N_0 N_1,$$
$$M_0 = N_0 - 1 - (b-1)\ell, \quad T_* = (B-2)\sum_j B^{H_j},$$
$$\mathcal{T} = M_0 + T_* N_0, \qquad \mathcal{J} = 2(1+g)^2\mathcal{D}(B) + B\sum_{v=0}^{K_*} B^v,$$
$$\mathcal{S} = g + \mathcal{J}N_0, \qquad R = (N-1)\big((\mathcal{S}+\mathcal{T}+1)N + \mathcal{T} + 1\big).$$

The source cap is $G_* = 32bB^W$ in the nine-variable case and $C_* = bB^W$ in the ten-integer case. All sums have fixed finite limits.

**Lemma 4.1** (The carry test survives the sparse encoding)**.** *The source has the same two-way interface as the source in package 45: if $b$ is a power of two and $0 < g < 4bB^W$, then*

$$N^2 \mid \binom{2R}{R} \quad \Longrightarrow \quad \exists \mathbf{x} \in \mathbb{N}^{31} \ \bigwedge_j r_j(\mathbf{x}) = 0.$$

*Conversely every common zero of the parent admits arbitrarily large power-of-two choices of $b$ with an encoding $0 < g < bB^W$ satisfying the central-binomial divisibility. The corresponding signed-source interface holds on the bounds required by the retained Sun bridge.*

*Proof.* Every coefficient of $\mathcal{D}$ is a single signed weighted coefficient, by Lemma 3.1; overlapping rows introduce no coefficient sum. Its leading coefficient is positive. If $B > M_* + 1$, the total absolute contribution of all lower positions at $B$ is smaller than one unit in the leading position, so $\mathcal{D}(B) > 0$. For $0 \leq x_i < b$, all coefficients of $\mathcal{C}$ are nonnegative and their sum is $(1 + \sum_i x_i)^2 \leq (32b)^2$. Therefore

$$|[T^v]\mathcal{C}(T)\mathcal{D}(T)| \leq M_*(32b)^2 < B/4.$$

Collisions among absent monomials do not change this sum bound. Adding $B/2$ at every position through $K_*$ yields digits strictly between $B/4$ and $3B/4$, hence between zero and $B$.

The elementary block estimates retained from the source proof give $0 \leq \mathcal{S}, \mathcal{T} < N$ on the soundness range $g < 4bB^W$. For example $1 + g \leq 4bB^W$, $g < N_0$, and

$$2(1+g)^2\mathcal{D}(B) < 32M_*b^2\frac{B^{K_*+1}}{B-1};$$

the choice of $\Lambda$ makes this smaller than the allowed high block. The geometric part of $\mathcal{J}$ is smaller than $2B^{K_*+1}$, so $\mathcal{J} < N_1$. Also $T_* < B^{H_*+1}$, giving $\mathcal{T} < N$. These estimates use $W, H_*, K_*$, not equal target spacing.

For $N = 2^r$ and $0 \le S, T < N$, the identity

$$(N-1)((S+T+1)N + T + 1) = (S+T)N^2 + (N-1-S)N + (N-1-T)$$

gives $\mathrm{pc}(R) = 2r - \tau_2(S,T)$, where $\tau_2$ is the number of binary carries with multiplicity. Legendre's factorial formula gives $v_2\binom{2R}{R} = \mathrm{pc}(R)$. Thus the divisibility is equivalent to no binary carries between $\mathcal{S}$ and $\mathcal{T}$.

The low-block mask first forces $g = \sum_i x_i B^{w_i}$ with $0 \le x_i < b$. The high-block mask can be divided by two together with $\mathcal{J}$, which is a common binary shift. At each selected position it tests all lower bits of the shifted digit. Within $(B/4, 3B/4)$ the only multiple of $B/2$ is $B/2$ itself. By Lemma 3.1, this is equivalent to the vanishing of every residual. Other positions need no additional condition.

Conversely choose $b$ larger than all coordinates of a common parent zero. Its encoding is positive because the all-zero tuple is not a parent zero; it is smaller than $bB^W$. Both masks are then disjoint from their encoded blocks. The signed source chooses arbitrarily large even powers of two $b = 1 + 3f$ and applies the same bounds. Its congruence requirement $3 \mid R$ follows from $B \equiv N \equiv 1 \pmod 3$, since the exponent 2140030 is even. These are the source conditions used by the inherited ten-integer arithmetic theorem. □

For arbitrary nine-variable outer assignments, positivity of $\mathcal{D}(B)$ and the masks also gives $R \ge N \ge 8$ and $0 < b \le R, N$. These hold before a putative solution is assumed. This order matters when invoking the arithmetic bridge to establish the eventual bound on $g$.

## 5 The nine-variable equation

Initially use mixed domains

$$\epsilon, \phi, n \in \mathbb{N}, \qquad g, h, s, w, i, j \in \mathbb{Z}_{>0}.$$

There are exactly nine unknowns. Using the source $R, N, b, g, G_*$, define the unchanged Jones registers

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M &= RY,\\
P &= 2M^2 U, & K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R + 1, & W_1 &= bw, & C &= B_1 + W_1 + \phi,\\
J &= 4A - 5, & Q &= 3W_1 C - 2(W_1^2 - 1), & &\\
D &= (A^2-1)C^2 + 1, & e &= iJC, & E &= CDe,\\
F &= (A^2-1)E^2 + 1, & G &= A + F(CD + 1 - A), & &\\
H &= B_1 + jC, & I &= (G^2-1)H^2 + 1, & &\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI, & &\\
u &= JF, & v &= QF + J(H - C), & &\\
m &= \big(K^2 - 4(C - KY)^2\big)G_* - 8gK^2, & M_d &= Q + H. & &
\end{aligned}$$

The quantity $m$ is a polynomial register, not another witness. The numerical Jones ratio theorem says that the existential conditions

$$a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}), \qquad u \mid v, \qquad m > 0 \tag{5}$$

recover the central-binomial divisibility, a power-of-two base, and $8g < G_*$. Conversely the source's stronger $32g < G_*$ supplies these conditions with appropriate arithmetic witnesses. We retain the precise `ratio_sound` and `ratio_complete` theorems, not an unmodified theorem with the older exponential source positions.

### 5.1 A smaller separating multiplier

Put $B_* = P(P+R)$. The source bounds imply $U \geq 64$, $M \geq 512$, and $A \geq 33280$. In particular all the following inequalities hold for *every* assignment in the specified domain, before imposing any square or divisibility condition:

$$0 < \sqrt{a_0} < PK \leq hB_* < h\sqrt{a_1}. \tag{6}$$

Indeed, $h(P+R) - K = (h-1)(R+1) \geq 0$, and $a_0 = P^2K^2 - K^2 + 1 < P^2K^2$. Also

$$A^2 - 2P = M^2(U-1)^2 > 0, \qquad D > A^2 > 2P, \qquad E > D.$$

It follows that $F > 2P^2 > B_*$, since $R < P$. The quantity $L = CD + 1 - A$ is greater than one, so $G > F > B_*$. Finally $I \geq G^2$ and $DF > 1$, proving the last bound in (6). Thus the existing coordinate $h$ already separates the two radicals. No new unknown is needed for that purpose.

### 5.2 The principal polynomial majorant

Define $\lambda = 2048$ and the following polynomial registers:

$$\begin{aligned} L &= CD + 1 - A, & \mathcal{N} &= 2G + L, \\ \mathcal{N}_1 &= 2e^2D(D-1)L + 3C, & \mathcal{N}_2 &= 2e^2(D-1)L, \\ \mathcal{N}_3 &= 2e^2(L-C), & \mathcal{N}_4 &= 2e^2C, \\ c_* &= 256i(e^2+2) + 2048, & B_g &= 2P(P+R) + 1. \end{aligned}$$

Put

$$T = e\big((1024D^2 - 512D - 128)\mathcal{N} - 64\mathcal{N}_1 - 40\mathcal{N}_2 - 28\mathcal{N}_3 - 21\mathcal{N}_4\big) + c_*, \tag{7}$$

$$V = TH + B_g, \qquad b_1 = \lambda^2 a_1. \tag{8}$$

These definitions have no division. The registers $\mathcal{N}_k$ are explicit products and sums, not quotient operations.

**Lemma 5.1** (Uniform polynomial majorant)**.** *For every allowed assignment,*

$$T > \lambda\sqrt{DF}\,G > 0, \qquad \sqrt{b_1} < V - B_*, \qquad B_* < \sqrt{b_1}.$$

*Proof.* It is enough for the first inequality that $A \geq 3$ and $C, i \geq 1$. Let $X_* = e^2D^3(D-1)$. Direct substitution gives

$$DF = X_* + D, \qquad G = (X_*/D)L + CD + 1.$$

The fractions in this proof are real-number comparisons only. Define the two integer polynomials

$$\begin{aligned} t_6(D) &= 1024D^6 - 512D^5 - 128D^4 - 64D^3 - 40D^2 - 28D - 21, \\ d_5(D) &= 33792D^5 + 10560D^4 + 4928D^3 + 2464D^2 + 1176D + 441. \end{aligned}$$

Their exact identity is

$$t_6(D)^2 - 1024^2D^{11}(D-1) = d_5(D). \tag{9}$$

For $D \geq 1$, $t_6(D) > 0$: the sum of the negative coefficients is 793, less than 1024, and every corresponding power is at most $D^6$. Also $d_5(D) > 0$. Hence, for $D > 1$,

$$\frac{t_6(D)}{D^4} > 1024D^2\sqrt{1 - 1/D}.$$

This finite identity replaces any need to assume convergence of a series.

The exact remainders $\mathcal{N} = D^k \mathcal{N}_k + r_k$ are

$$r_1 = 3 - A, \qquad r_2 = 3CD + 3 - A,$$
$$r_3 = 2e^2(A-1)D^2 + r_2, \qquad r_4 = 2e^2(1 - A - C)D^3 + r_3.$$

They give the identity

$$T = e\frac{t_6(D)}{D^4}\mathcal{N} + \mathcal{E} + c_*, \qquad \mathcal{E} = e\left(\frac{64r_1}{D} + \frac{40r_2}{D^2} + \frac{28r_3}{D^3} + \frac{21r_4}{D^4}\right).$$

Since $D \geq A^2 \geq 9$ and $e = i(4A-5)C$, elementary bounds give

$$\frac{eA}{D} < \frac{9}{2}i, \qquad \frac{eC}{D} < \frac{3}{2}i, \qquad \frac{e(A+C-1)}{D} < 6i.$$

For example, the first follows from $eA/D < 4iA^2/((A^2-1)C) \leq 9i/2$; the second follows from $eC/D < 4iA/(A^2-1) \leq 3i/2$. Both $r_2$ and $r_3$ are positive. Bounding only the possibly negative terms therefore yields

$$\begin{aligned} -\mathcal{E} &< \frac{64eA}{D} + \frac{42e^3(A+C-1)}{D} + \frac{21eA}{D^4} \\ &< 252ie^2 + \frac{765}{2}i < c_* - 2048. \end{aligned}$$

For positive $X_*, D$, strict concavity of the square root gives $\sqrt{X_* + D} < \sqrt{X_*} + D/(2\sqrt{X_*})$. Multiplication by $2048G$ and the displayed identity for $G$ give

$$\lambda\sqrt{DF}\,G < 1024eD^2\sqrt{1 - 1/D}\,(2G + L) + \lambda v_*, \quad v_* := \frac{CD+1}{2eD\sqrt{1-1/D}} < 1.$$

The last inequality follows from $e \geq 7C$ and $\sqrt{D(D-1)} > D - 1$. Combining these bounds proves $T > \lambda\sqrt{DF}\,G$. Since $I \leq G^2H^2$, equations (8) imply $V - \sqrt{b_1} > B_g > B_*$. The remaining lower bound follows from (6). □

The low-degree padding $B_g = 2B_* + 1$ has two roles. It supplies the strict root gap for all inputs, and its factor two prevents the highest terms of the final norm from cancelling. Omitting that factor preserves the gap but invalidates the stated exact-degree calculation.

### 5.3 The final equation and its exact existential meaning

Set

$$\Omega = um,$$
$$\Theta = u(n + M_d) + v - \Omega(2M_d + hV).$$

The principal nine-variable equation is

$$\boxed{F_9 = \left(\Theta^2 - \Omega^2(a_0 + h^2 b_1)\right)^2 - 4\Omega^4 h^2 a_0 b_1 = 0.} \tag{10}$$

Together with the displayed source, register definitions, fixed coefficient bytes, weights, and targets, this is a completely specified ordinary integer polynomial in exactly nine unknowns.

**Proposition 5.2** (No lost sign or margin condition)**.** *For fixed values of the first eight mixed-domain coordinates,* (10) *admits* $n \in \mathbb{N}$ *if and only if* $a_0, a_1$ *are integer squares,* $u \mid v$*, and the computed margin* $m$ *is positive.*

*Proof.* Here $u > 0$ and $0 < v/u < M_d - 1$. The latter follows directly from $v/u = Q/J + (H - C)/F$, with $Q > 0$, $J, F > 1$, and $H - C > 0$. If $m = 0$, the left-hand polynomial is $(u(n + M_d) + v)^4 > 0$. Thus a zero has $\Omega \neq 0$. Factoring the four-root norm over the real numbers gives

$$\frac{\Theta}{\Omega} = \varepsilon\sqrt{a_0} + \delta h\sqrt{b_1}, \qquad \varepsilon, \delta \in \{-1, 1\}.$$

This rational sum is nonzero by (6). If it is $r$, then

$$\sqrt{a_0} = \frac{r^2 + a_0 - h^2 b_1}{2\varepsilon r}.$$

Both roots are therefore rational and hence integral. This handles equal or dependent square classes as well. Because $b_1 = \lambda^2 a_1$, integrality of $\sqrt{b_1}$ implies integrality of $\sqrt{a_1}$: a rational square root of an integer is an integer.

Write the signed integer roots as $r_0, r_1$ and define

$$Z = 2M_d + hV + r_0 + hr_1.$$

The strict root bounds, with integer endpoints, give $r_0 \geq -hB_* + 1$ and $r_1 \geq -V + B_* + 1$. Consequently $Z \geq 2M_d + h + 1 > 0$. Rearranging the equation for $\Theta$ gives

$$n + M_d + v/u = mZ.$$

Its left side is positive, so $m > 0$. Its right side and $n + M_d$ are integers, so $u \mid v$.

Conversely, suppose the three conditions hold. Choose either signs of the integral roots, retain the *actual* positive computed value of $m$, and set $n = mZ - M_d - v/u$. The same bounds imply $n \geq 0$, and substitution makes (10) vanish. No step replaces the computed margin by a freely chosen witness or assumes $m = 1$. □

Composing Proposition 5.2 with the numerical ratio bridge and the source lemma proves the nine-variable existence equivalence.

### 5.4 A smaller-cost alternative

The release also supplies the simpler choice $\lambda = 8$,

$$S_8 = e(8D^2 - 4D - 1), \qquad V_8 = S_8 GH, \qquad b_1 = 64a_1.$$

Use the same norm (10) with $V = V_8$. Its exact defect is

$$V_8^2 - 64a_1 = (e^2(8D + 1) - 64D)G^2H^2 + 64DF(H^2 - 1).$$

Since $e \geq 7$, the first bracket is greater than $4De^2$. Also $V_8 < 8D^2 eGH$ and $2B_* < D^2$, so

$$\frac{V_8^2 - 64a_1}{V_8} > \frac{eGH}{2D} > \frac{(D - 1)De^3}{2} > D^2 > 2B_*.$$

As $V_8 > B_*$, this implies $(V_8 - B_*)^2 > 64a_1$ and proves the same strict gap. The preceding norm proof applies unchanged. This alternative uses 568 operations with uniform targets or 624 with overlapping targets; its exact degrees are respectively 24,399,300 and 18,262,476. It is included because operation count and degree have different minima among the supplied constructions.

For nine strictly positive coordinates, replace $\epsilon, \phi, n$ by their positive coordinates minus one and leave the other six unchanged. For nine nonnegative coordinates, leave those three unchanged and replace $g, h, s, w, i, j$ by their nonnegative coordinates plus one. These are exact domain bijections. The emitted circuits charge all shifts and use the same fixed coefficient tuple.

## 6 The ten-unrestricted-integer equation

This branch uses $(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}$ and the signed source $b = 1 + 3f$, $B = \Lambda b^2$, $X = R$, $Y = N^2$, $C_* = bB^W$. The following registers are retained from the fully explicit package-40 Sun adapter:

$$\begin{aligned}
&L = lY, && U = 2LX, && V = 4gwY, && W_0 = bw,\\
&A = U(V+1), && B_1 = 2X+1, && C = B_1 + (A-2)h,\\
&P = U^2V, && K = X + 1 + k(P-2),\\
&S_0 = 2A - 5, && T_0 = 3W_0C - 2(W_0^2 - 1).
\end{aligned}$$

$$\begin{aligned}
r_1 &= (P^2-4)K^2 + 4,\\
\delta &= 8C_*^3gK^2 - g^2\big(32(C-KL)^2C_*^3 + g^2K^2\big).
\end{aligned}$$

$$\begin{aligned}
&O = f^2l^2\delta, && \xi = zx,\\
&D = (A^2-4)C^2 + 4, && E = C^2D\xi,\\
&F = 4(A^2-4)E^2 + 1,\\
&G_0 = 2(A+2)(A-2)^2E^2 - 1, && \Omega_0 = 16E,\\
&H_0 = C + (B_1 - C)F + \Omega_0Fy,\\
&I_0 = (G_0^2 - 1)H_0^2 + 1.
\end{aligned}$$

Put

$$\begin{aligned}
&\Sigma = \xi S_0^5, && r = T_0 + S_0y,\\
&\vartheta = (r^3 - rS_0^2 - S_0^3)(r^2 + 23S_0^2),\\
&r_0 = b\big(1 + (3O-4)b^2z^2\big), && r_2 = DFI_0,\\
&w_0 = 1, && w_1 = r_0, && w_2 = r_0(P^2 + K^2 + 3),\\
&Z = \Sigma m - \vartheta, && A_i = \Sigma^2 w_i^2 r_i \quad (0 \le i \le 2),\\
&V_* = Z^2 + A_2 - A_0 - A_1, && W_* = V_*^2 + 4Z^2A_2 - 4A_0A_1.
\end{aligned}$$

The equation is

$$\boxed{F_{10} = W_*^2 - 16V_*^2Z^2A_2 = 0.} \tag{11}$$

No square root, division, binomial coefficient, or variable exponent occurs in its definition. Those objects occur only in the correctness proof.

### 6.1 What is reused in this branch

The retained finalizer is the product over eight sign choices of a three-radical expression, expanded into the displayed norm. Its proof recovers the square predicates and merged divisibility condition using separating weights, including dependent square classes. The case $\Sigma = 0$ yields $\vartheta^8 \neq 0$ and is excluded in the inherited argument. The polynomial $(t^3 - t - 1)(t^2 + 23)$ supplies the cited divisibility–nonvanishing merger, with full residue interpolation from Sun's theorem [7, 2].

We replace only its source. Section 4 verifies the source's positivity, range, base, congruence, and carry requirements for the new positions. The finalizer first establishes the hypotheses needed by that source; a square base alone is never treated as a power of two. The inherited signed-tail theorem then proves that (11) has an integer zero exactly when the quadratic parent has a positive zero. This branch's new contribution is the certified sparse and overlapping source, not a new proof of the general ten-integer theorem.

## 7 Why the composition is a ZFC consistency equivalence

The complete mathematical chain is

$$\begin{aligned} \text{new polynomial has a zero} &\iff \text{the encoded quadratic system has a zero} \\ &\iff \text{the literal K215 machine halts on blank tape} \\ &\iff \text{ZFC has a finite syntactic refutation.} \end{aligned}$$

The first equivalence is the new encoding proof composed with the explicitly retained numerical bridge. The latter two equivalences are the inherited native-window/Pell compiler and the repository's calibrated proof enumerator. They are substantive mathematical dependencies, not conclusions of running the finite checkers in this archive.

The new arguments concern finite polynomials, integer arithmetic, finite coefficient lists, and elementary algebraic-number reasoning. They can be formalized in ZFC. Combining them with the identified inherited proofs therefore supplies the asserted ZFC-provable equivalences. No assertion of the general soundness of Lean or Rocq inside ZFC is used as a replacement for this proof chain.

### 7.1 A subtle inherited rank lemma checked in this review

The parent uses $I = ic^2$, $I^2 = \Delta(f^2 - 1)$, rather than the stronger-looking expression with an additional $\Delta$ in $I$. This must not be justified by simply cancelling $\Delta$, because $\Delta$ need not be squarefree. The inherited relaxed-rank lemma handles it correctly. In the notation $\Delta = A^2 - 1$, $c = \psi_A(p) > A\Delta^2$, it establishes

$$f = \chi_A(m), \quad p \mid m, \quad c \mid m, \quad ic^2 = \Delta\psi_A(m).$$

Its proof passes to the fundamental unit of the squarefree part of $\Delta$, applies strong divisibility, and uses the independently obtained large-index bound to force the correct index divisibility. The full audit and source locator are included in `proofs/FOUNDATION_AUDIT.md`. Thus this parent does not silently assume $\Delta$ squarefree.

### 7.2 Formalization status

At the reviewed commit, the repository has an unconditional external Lean theorem for a nine-natural-variable polynomial selected via `Classical.choose`, and a 58-natural-variable quartic. Those

are different objects from the materialized coefficient circuits here. The active work log records an executable coefficient compiler and an object-language ZFC derivation as open formalization tasks [5]. The short explicit P67/P69/P73/P75 families have local Lean algebraic certificates, with their end-to-end native compiler bridge still an ordinary mathematical proof. This contribution adds ordinary proofs and independently checked finite data; it does not claim to close those formalization tasks.

No consistency assumption is used to define the coefficients. The theorem proves an equivalence and proves neither side. A claim of two-sided independence would require additional hypotheses and is not made here.

## 8 Reproduction and the scope of the evidence

The release contains Python code using only the standard library. The recommended entry point is `verify.py`; `verify_refined.py` checks the refined nine-variable circuits. The independent support checker does not import construction helpers. The scripts are also run with `python3 -O`, so correctness gates must not disappear when assertions are disabled. The archive includes exact receipts and a complete hash manifest.

The checks establish the following finite claims:

1. The parent expansion and all successor shifts match the fixed 31-coordinate, 22-equation source.
2. All 71 present monomials are isolated against all 528 possible quadratic monomials, and every cross-row target is isolated in both layouts.
3. The four compiler ports and all derived coefficient recipes are bound to the supplied bytes; source positivity and coefficient bounds use that complete tuple.
4. The emitted straight-line graphs are acyclic, every gate and witness port is live, and every operation is charged under the stated model.
5. Direct source-formula replay matches the emitted graphs; exact symbolic identities relate the finalizer to its cancellation-resolved degree certificate.
6. Multivariate degree upper bounds are attained, by a nonzero diagonal leading coefficient, with exact formulas for the nine-variable tail and independent modular certificates for the ten-variable tail.

These are checks of the polynomial construction and its finite certificates. Searching sample assignments cannot prove an unbounded consistency equivalence. The universal claims instead follow from the mathematical proofs above and their named inherited dependencies.

## 9 Where the reduction comes from and what remains

The first gain comes from specializing the algebraic encoding to the *support* of the actual residual system. The 457 absent quadratic monomials never need to be separated from one another. This is stronger than merely searching for a better full Sidon set, because it changes the constraint being solved.

The second gain separates only the output positions that will actually be tested. The row-polynomial supports may overlap freely elsewhere. The explicit target list is a small exact certificate for this weaker condition, while every carry bound is preserved.

The third gain uses divisibility already present in the arithmetic tail to approximate its radical more closely by a polynomial. It also replaces a large separating multiplier by the existing witness $h$,

after proving an unconditional gap. These transformations introduce no witness and no variable denominator. Their defect identities explain the lower degree and provide small exact certificates of the otherwise hidden cancellations.

Further work can optimize the joint choice of variable weights, row order, and charged power schedule; formalize Lemma 3.1 once for all finite sparse sources; and connect executable coefficient data to the existing Lean ZFC endpoint. None of those possible improvements is included in the certified numerical claims of this release.

# A Exact finite placement data

The overlap table gives the original parent row number, not its index in the reordered construction array.

| Target index $j$ | $H_j$ | Original row | Sign |
|---:|---:|---:|---:|
| 0 | 358 | 14 | +1 |
| 1 | 608 | 7 | +1 |
| 2 | 939 | 21 | +1 |
| 3 | 1147 | 10 | +1 |
| 4 | 1432 | 8 | +1 |
| 5 | 1739 | 17 | +1 |
| 6 | 1989 | 16 | +1 |
| 7 | 2170 | 6 | +1 |
| 8 | 2512 | 1 | +1 |
| 9 | 2699 | 13 | +1 |
| 10 | 2972 | 4 | +1 |
| 11 | 3305 | 2 | +1 |
| 12 | 3492 | 19 | +1 |
| 13 | 3683 | 11 | +1 |
| 14 | 3914 | 20 | +1 |
| 15 | 4108 | 5 | +1 |
| 16 | 4399 | 18 | +1 |
| 17 | 4697 | 15 | +1 |
| 18 | 4926 | 22 | +1 |
| 19 | 5224 | 3 | +1 |
| 20 | 5469 | 12 | +1 |
| 21 | 5775 | 9 | −1 |

The uniform placement uses all positive signs and original row order $1, 2, 3, 4, 6, 7, \ldots, 22, 5$, with $H_j = 358 + 359j$. Both layouts use the weights in equation (2).

# B Detailed degree certificate

Let $t = \deg N = 3(H_* + 3W + 2)$. Thus $t = 25\,308$ for uniform targets and $t = 18\,942$ for overlapping targets. All degrees below are multivariate total degrees. A *leader* is the leading coefficient after substituting the same indeterminate for all nine coordinates. A nonzero such coefficient proves that

the multivariate upper bound is attained. Successor/predecessor domain shifts leave these leaders unchanged.

| Register | Degree | Diagonal leader |
|---|---|---|
| $N$ | $t$ | $2^2$ |
| $R$ | $3t$ | $2^4$ |
| $P$ | $12t+3$ | $2^{21}$ |
| $A$ | $7t+2$ | $2^{12}$ |
| $C$ | $3t$ | $2^5$ |
| $D$ | $20t+4$ | $2^{34}$ |
| $e$ | $10t+3$ | $2^{19}$ |
| $L$ | $23t+4$ | $2^{39}$ |
| $F$ | $80t+18$ | $2^{140}$ |
| $G$ | $103t+22$ | $2^{179}$ |
| $H$ | $3t+1$ | $2^5$ |
| $a_0$ | $48t+14$ | $2^{84}$ |
| $a_1 = DFI$ | $312t+68$ | $2^{542}$ |
| $\Omega$ | $115t+3W+31$ | $-2^{211}$ |
| $T$, principal majorant | $153t+33$ | $2^{277}$ |
| $V$, principal majorant | $156t+34$ | $2^{282}$ |

### B.1 A polynomial certificate for the hidden cancellation

Use $t_6, d_5, r_k, \mathcal{N}, L, c_*$ from the majorant proof and define

$$R_* = e(64r_1D^3 + 40r_2D^2 + 28r_3D + 21r_4) + c_*D^4.$$

Then $D^4T = e\,t_6\mathcal{N} + R_*$, and the following are exact integer-polynomial identities:

$$\begin{aligned}\Delta_T &:= D^8(T^2 - \lambda^2 DFG^2)\\ &= e^2 d_5 \mathcal{N}^2 + 1024^2 D^8(X_* L^2 - 4DG(CD+1))\\ &\quad + 2e\,t_6\mathcal{N}R_* + R_*^2, \qquad (12)\\ \Delta_V &:= D^8(V^2 - \lambda^2 a_1)\\ &= H^2\Delta_T + \lambda^2 DFD^8(H^2-1) + D^8(2THB_g + B_g^2). \qquad (13)\end{aligned}$$

Here $X_* = e^2D^3(D-1)$. The term $X_*\mathcal{N}^2 - 4(X_*+D)G^2$ simplifies to $X_*L^2 - 4DG(CD+1)$; this is the essential cancellation in (12).

The respective degree bounds of the four summands in (12) are

$$326t+70, \quad 306t+62, \quad 333t+72, \quad 200t+46.$$

For the third bound, $\deg R_* = 100t+23$, with a positive leading coefficient supplied by $c_*D^4$. Its term uniquely dominates. Since $\deg D^8 = 160t+32$, we have

$$\deg(T^2 - \lambda^2 DFG^2) = 173t+40.$$

After multiplication by $H^2$, that becomes $179t+42$. In $V^2 - \lambda^2 a_1$, however, the padding cross term $2THB_g$ has degree $180t+40$, which is greater because $t > 2$. Its leader is $2 \cdot 2^{282} \cdot 2^{43} = 2^{326}$. All other terms have smaller degree. Thus

$$\deg(V^2 - \lambda^2 a_1) = 180t+40, \qquad \mathrm{lc}_{\mathrm{diag}}(V^2 - \lambda^2 a_1) = 2^{326}.$$

Let $E_0 = u(n + M_d) + v - 2\Omega M_d$. The inner norm has the exact expansion

$$\Theta^2 - \Omega^2(a_0 + h^2 b_1) = E_0^2 - 2E_0\Omega hV + \Omega^2 h^2(V^2 - b_1) - \Omega^2 a_0.$$

The third term uniquely dominates: its degree is $410t + 6W + 104$ and its diagonal leader is $2^{748}$. Squaring produces degree $820t + 12W + 208$ and leader $2^{1496}$. The subtracted norm term has the *same* degree and leader $2^{1494}$. Therefore

$$\boxed{\deg F_9 = 820t + 12W + 208, \qquad \mathrm{lc}_{\mathrm{diag}}\, F_9 = 3 \cdot 2^{1494}.}$$

This proves exact degree, not merely a syntactic upper bound. It also explains the doubled padding $B_g$: padding only by $B_* + 1$ would make the two highest terms cancel.

### B.2 What the certificate graph evaluates

The primary graph evaluates exactly $F_9$. The principal degree certificate graph instead evaluates $D^{16}F_9$, through

$$\left[D^8(E_0^2 - 2E_0\Omega hV - \Omega^2 a_0) + \Omega^2 h^2 \Delta_V\right]^2 - 4D^{16}\Omega^4 h^2 a_0 b_1.$$

This avoids division while exposing every required cancellation. Its exact degree is $1140t + 12W + 272$ and its diagonal leader is $3 \cdot 2^{2038}$. Subtracting the known degree of $D^{16}$ recovers the primary degree. The certificate graph is not substituted for the delivered equation.

The V8 graph uses an equal-polynomial certificate instead. Its displayed defect has degree $252t + 56$ and diagonal leader $2^{443}$. Consequently

$$\deg F_{9,\mathrm{V8}} = 964t + 12W + 240, \qquad \mathrm{lc}_{\mathrm{diag}}\, F_{9,\mathrm{V8}} = 2^{1730}.$$

### B.3 Unchanged-finalizer comparison and modular receipts

For the source replacements with the inherited finalizers, the degree formulas are

$$\deg F_{9,\mathrm{inherited}} = 3456H_* + 10380W + 7192,$$
$$\deg F_{10} = 6320H_* + 19016W + 12992.$$

The inherited-finalizer graphs and their nonzero leading coefficients are independently replayed modulo $1\,000\,000\,007$, $1\,000\,000\,009$, and $998\,244\,353$. The refined graphs use $1\,000\,003$, $1\,000\,033$, and $1\,000\,000\,007$. A single nonzero residue at a proved multivariate upper bound suffices for exactness; the three receipts provide additional cross-checks. For the principal nine-variable graph the symbolic degree calculation above also supplies an exact integer leader.

## C Files, fixed coefficients, and provenance

The machine-readable equations live in `generated/`. The primary refined nine-variable equations and their degree certificates live in `refined/`; the principal certificate evaluates $D^{16}F_9$, while the V8 certificate evaluates $F_9$ itself. `data/` contains the literal parent, complete ports, coefficient recipes, weights, and placement data. `proofs/` contains the independent mathematical reviews; `upstream/` and `inherited/` contain the pinned dependency text needed to inspect the inherited interfaces. The source

and dependency hashes are recorded in the manifests. The compiler ports are immutable data, never additional unknowns.

The manuscript source is `report.tex`; a normal TeX installation can rebuild it with two passes of `pdflatex`. The adjacent small TeX input files contain the exact tables and the finalizer proof. They are part of the source deliverable. New material is supplied under the repository's MIT No Attribution license; inherited notices and attribution remain in the archive.

## References


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[3] ProveIt report package 45. *Simultaneous quadratic coding for ZFC consistency*, 23 October 2026. Pinned directory `con-zfc-diophantine-forms/45-multioutput`; especially the source proof and `MULTITARGET_NINE_PROOF.md`.

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